



Triggers at Datadog: What Are Trigger Queues and Why You Should Use Them

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Who we are

Datadog is a world-class data platform providing real-time insights.



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Airflow contributor



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100T+ events
/day

More than a hundred
trillion events a day

Customers over
33,000+



Employees over
8,100+



Engineering over
3000+



Airflow adoption



60+
teams using Airflow

3x
since last year

Contributions

ADP Airflow contributions

Add status update

Insights

Workflows 7

Prioritized backlog

Status board

Roadmap

Bugs 🐛

In review

My items

Merged PRs

+ New view

status:Done

25

View

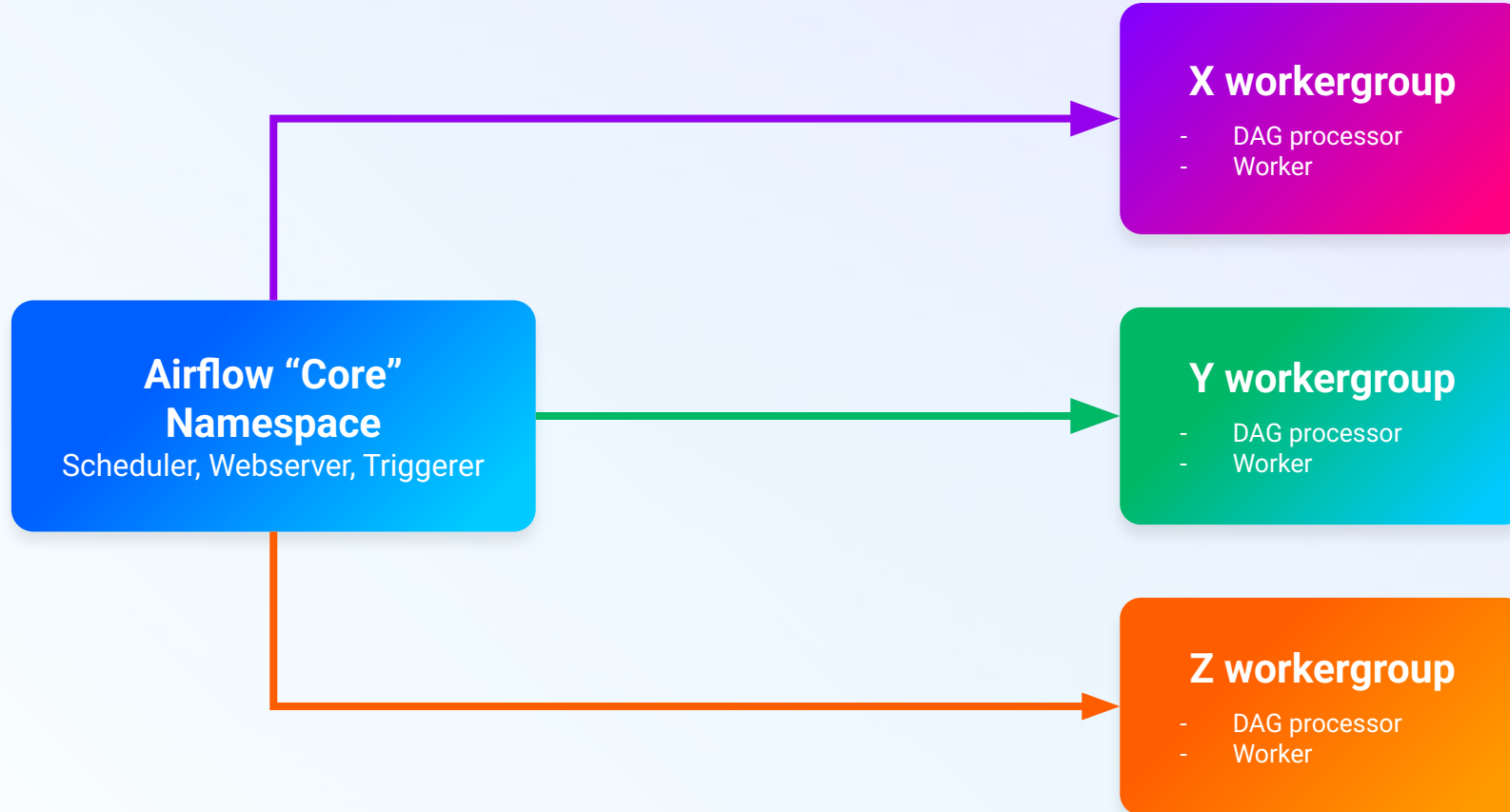
Title	Contributor	LOC	Created	Closed	Status
No Priority 25 Estimate: 0					
1 🔗 Enable triggerer queues #59239	zach-overflow	4842	Dec 9, 2025	Dec 31, 2025	Done
2 🔗 Optionally emit Dag tags as metric tags #68568	sortega	585	Jun 15, 2026	Jul 20, 2026	Done
3 🔗 Add trigger queue support for AsyncCallback and BaseEventTrigger #71346	zach-overflow	407	Aug 9, 2026	Aug 20, 2026	Done
4 🔗 Support a Unix Domain Socket for the metrics clients #70517	sortega	338	Jul 27, 2026	Aug 24, 2026	Done
5 🔗 Add overridable metadata engine creation hooks in settings.py #62184	avolant	232	Feb 19, 2026	Apr 1, 2026	Done
6 🔗 Add scheduler observability metrics #68068	safaehar	212	Jun 5, 2026	Jul 7, 2026	Done
7 🔗 Add deprecation warnings + doc for airflow.security.permissions #53796	zach-overflow	149	Jul 27, 2025	Jul 28, 2025	Done
8 🔗 Fix .airflowignore order precedence #56509	zach-overflow	127	Oct 9, 2025	Oct 18, 2025	Done
9 🔗 Add test to prevent ORM references in migration scripts #60176	zach-overflow	118	Jan 6, 2026	Jan 13, 2026	Done
10 🔗 Honor json_logs config in Celery worker startup #68916	hanxdatadog	111	Jun 23, 2026	Jun 29, 2026	Done
11 🔗 Don't deactivate DAG bundles owned by other dag-processors #69964	sortega	89	Jul 16, 2026	Jul 17, 2026	Done
12 🔗 Avoid scheduler crash when periodic maintenance actions fail #69487	hanxdatadog	59	Jul 6, 2026	Aug 7, 2026	Done
13 🔗 fix: guard the to.finish stats emission #67955	TimotheBarbe	46	Jun 3, 2026	Jun 5, 2026	Done

Shared-tenancy

- Centrally-managed by the platform team
- Teams run into each other
 - Deployment Clobbering
 - Noisy neighbor
- Compliance



Multi-tenancy solution - Workergroups



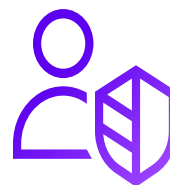
Initial workergroup model in 2025

Benefits of Workergroup



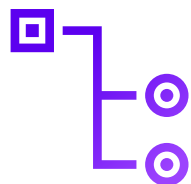
K8s resources

- Celery worker
- DAG processor



Namespace admin

- Self-managed RBAC
- Scaling



Deployment

- Custom rollout strategy
- Notification routing

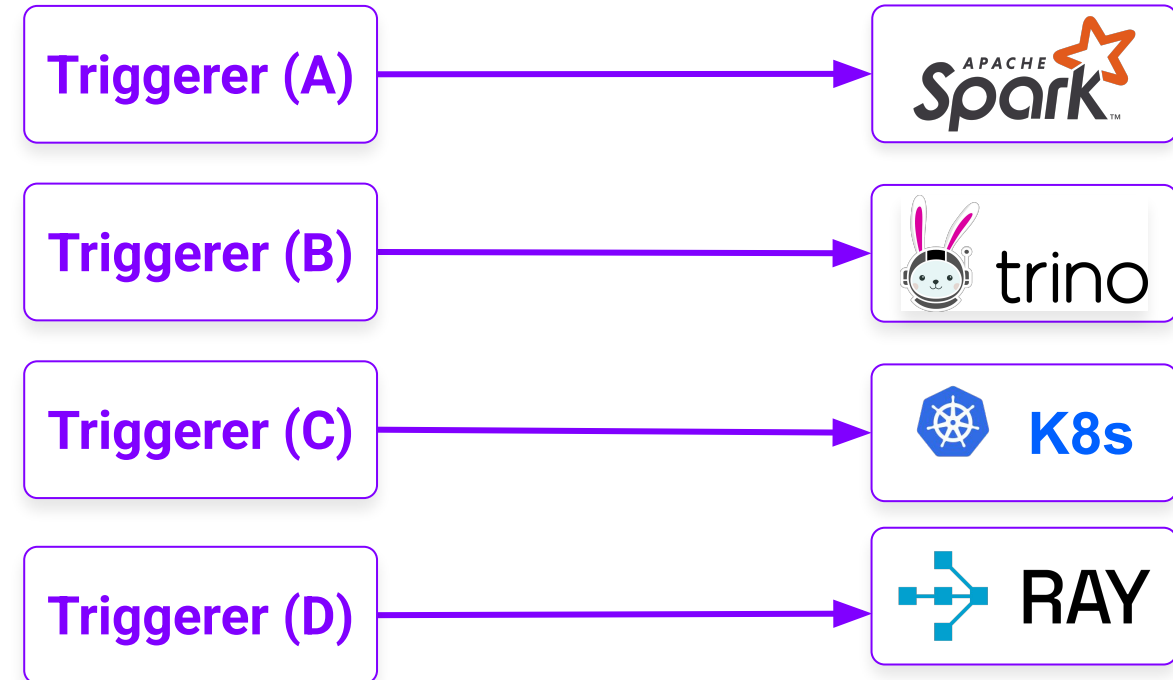


Cost attribution

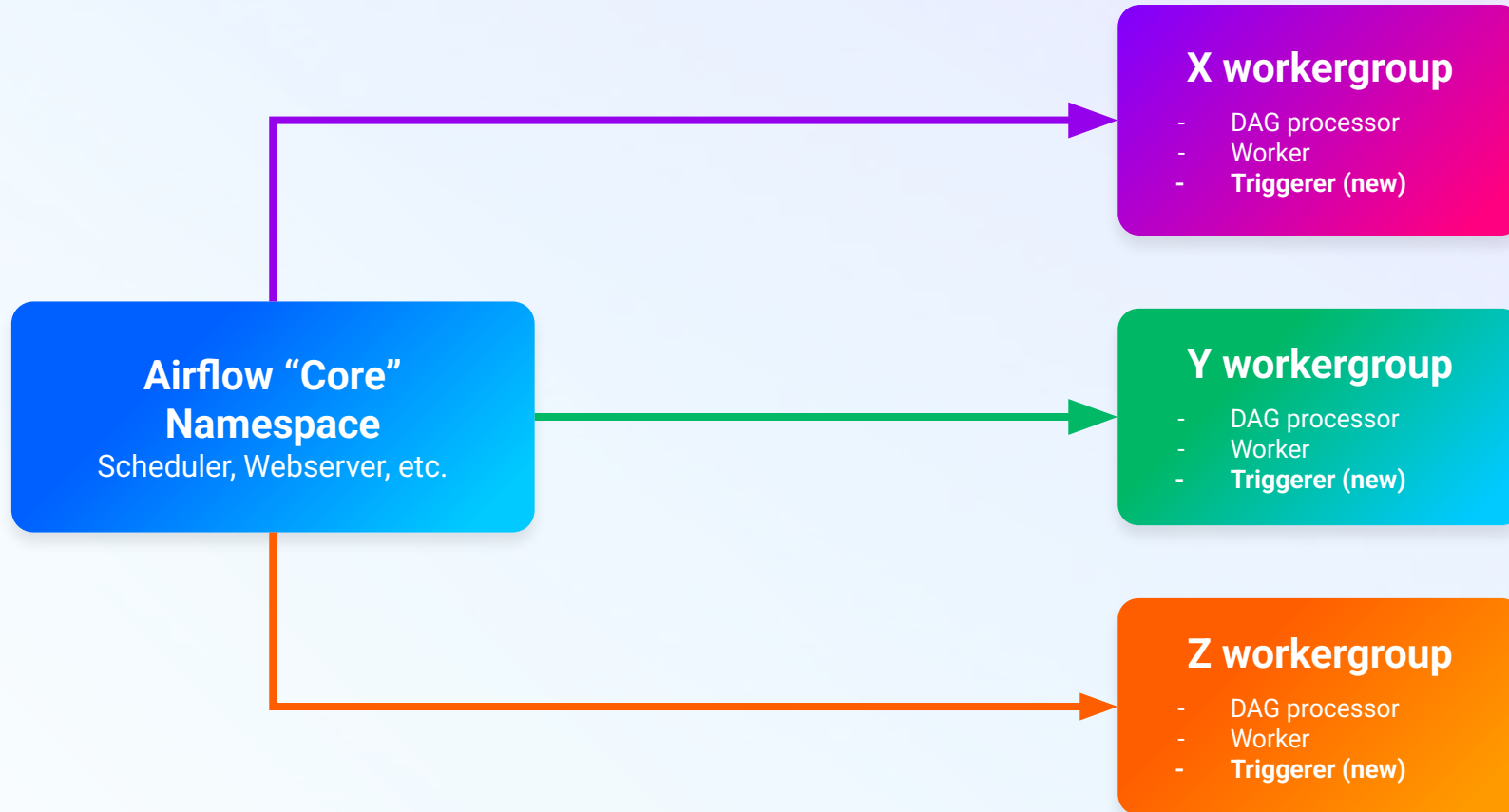
- Infra costs are attributed to individual teams

We need triggerer!

- Long running tasks
- External services



Worker groups now



Worker group model in 2026!

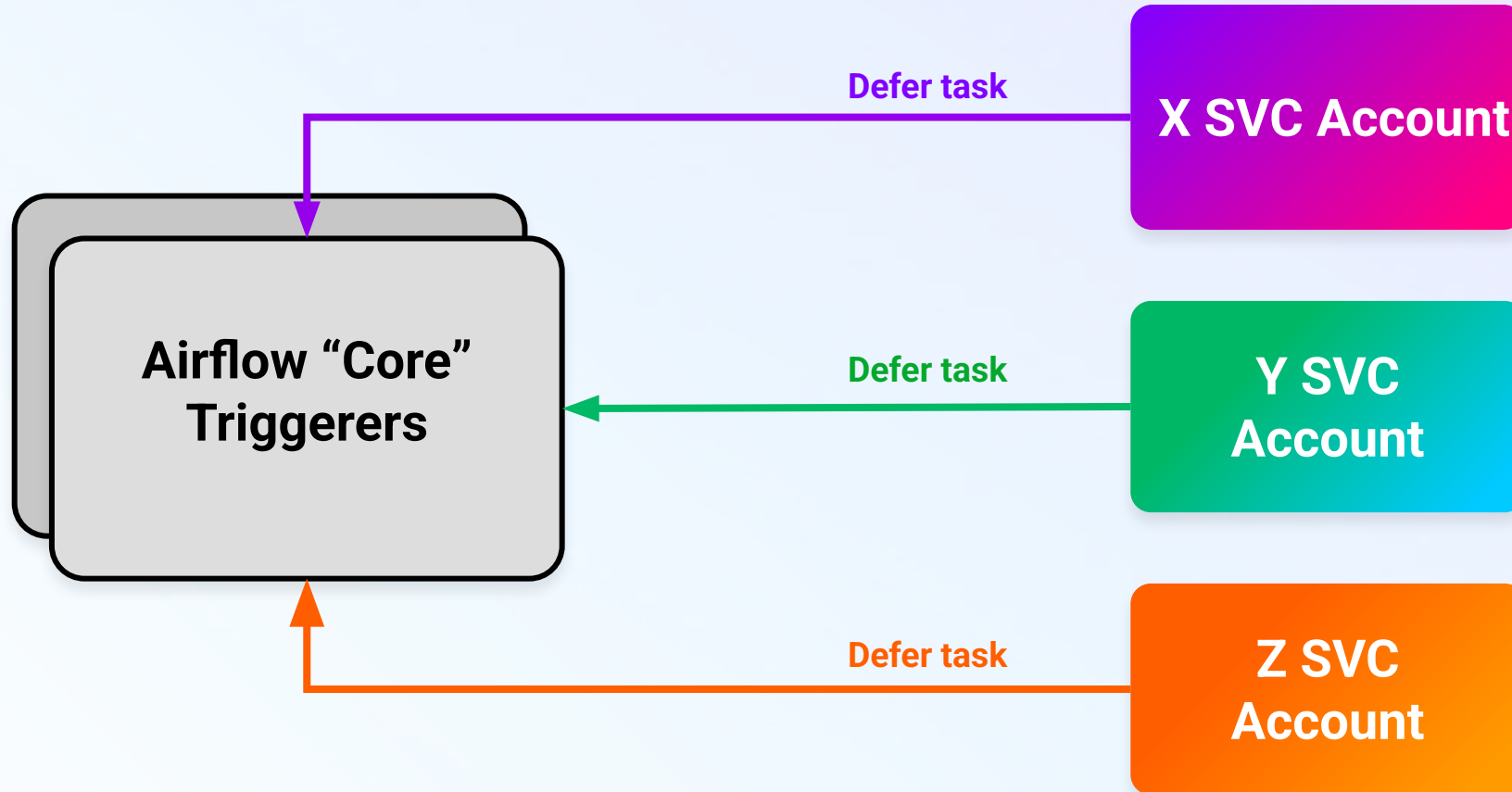
The Trigger Problem

Zach Gottesman

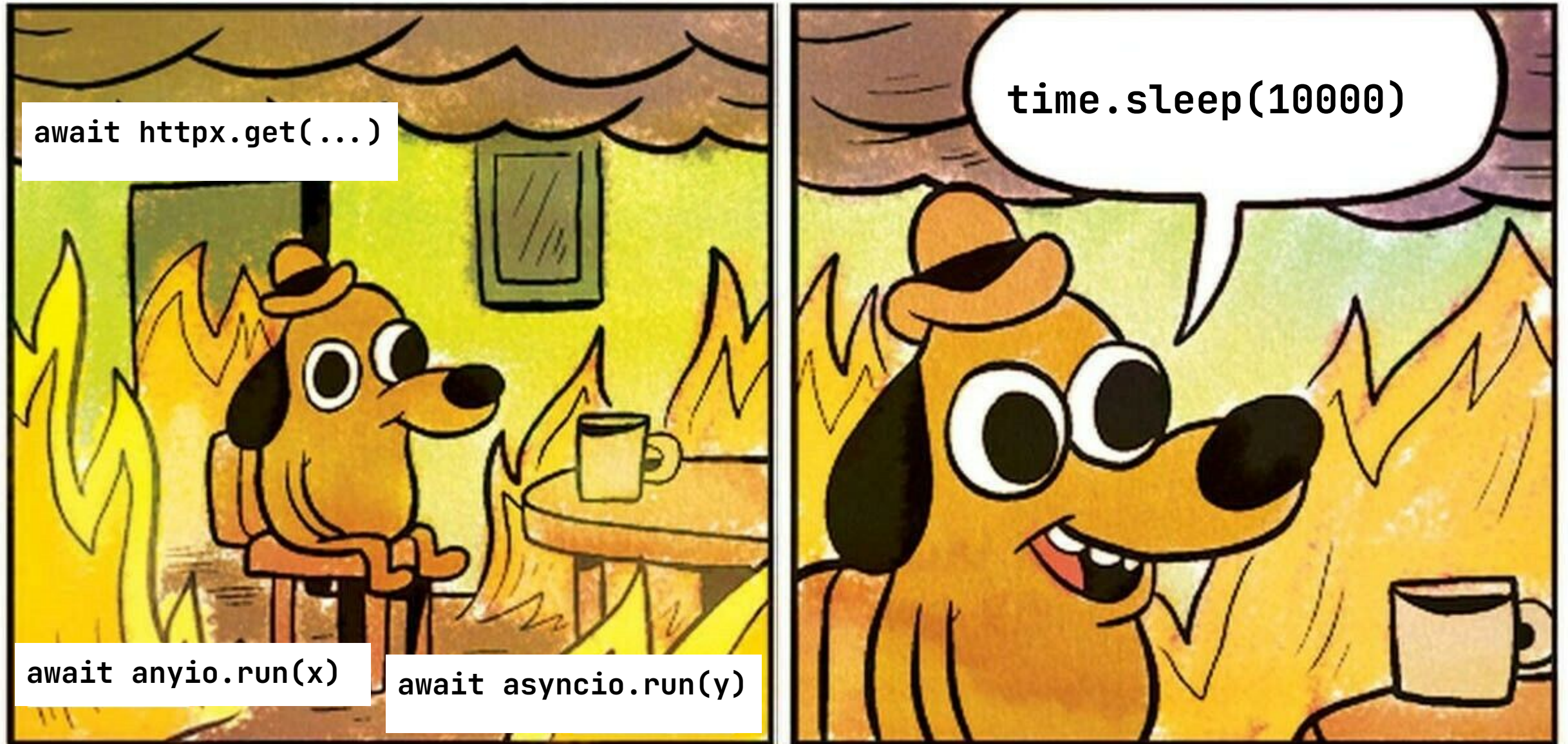
Senior Software Engineer, Datadog

*before Airflow 3.3.0

Before: Mismatched Identities



Before: Noisier Neighbors



Original artwork by KC Green,
gunshowcomic.com/648

Fig 4. Larger blast radius for blocking code

Mitigation Strategies?



Increased triggerer replicas?



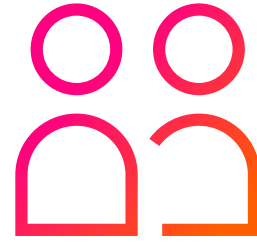
No team awareness



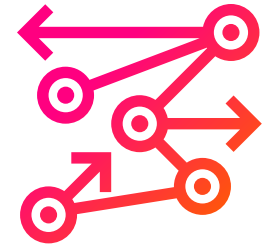
Increased cost



Custom “team-aware” triggers?



Sketchy role shadowing



Added complexity

Trigger Queues

github.com/apache/airflow/pull/59239

What Are Trigger Queues?

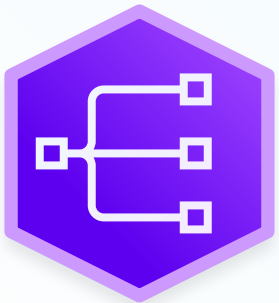


An opt-in feature for arbitrarily assigning Triggers to Triggerer nodes



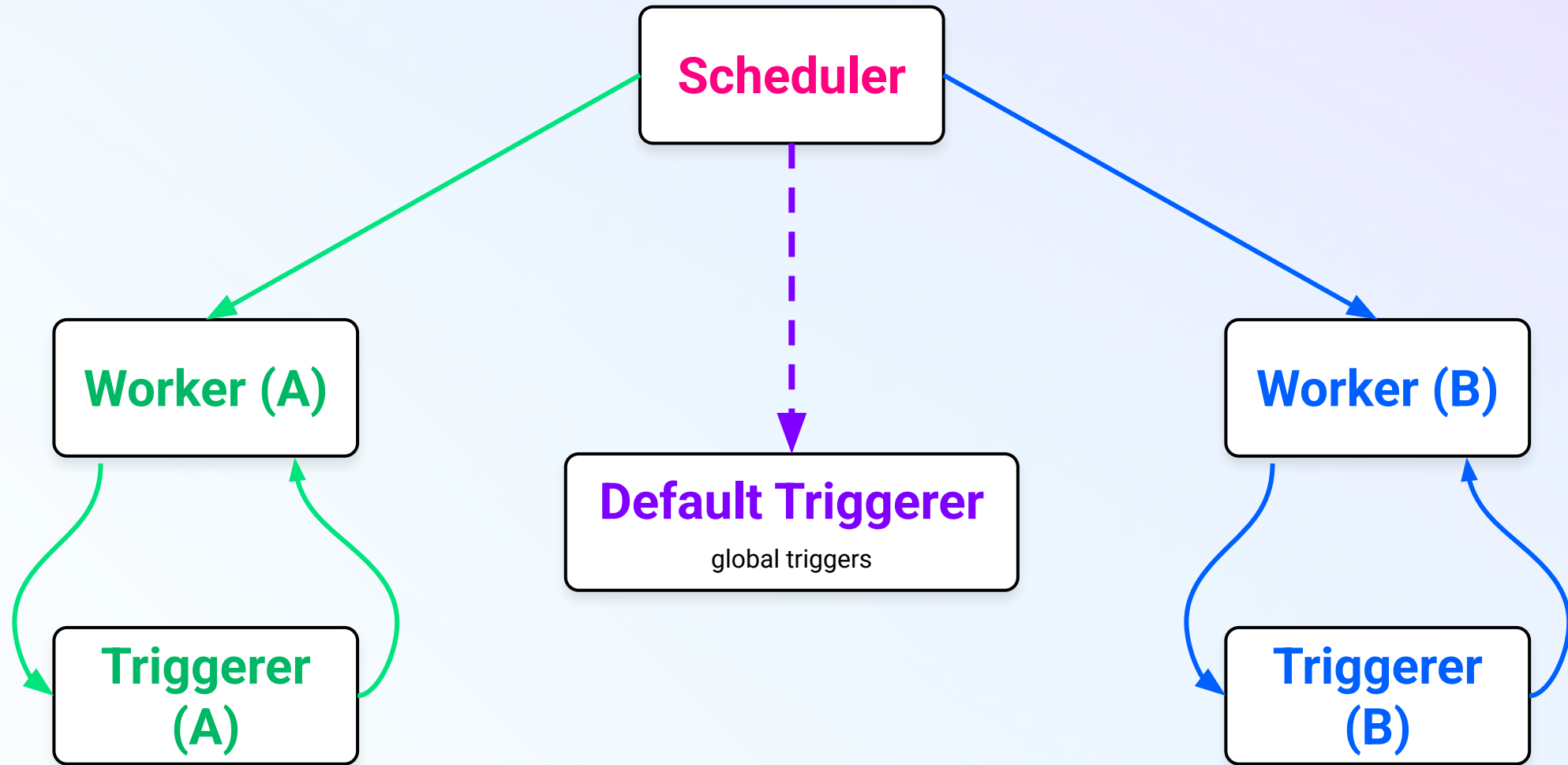
Follows from the task queue concept in Airflow

Worker pulls task from “Q” → Task defers → Trigger assigned to “Q” → Triggerer runs

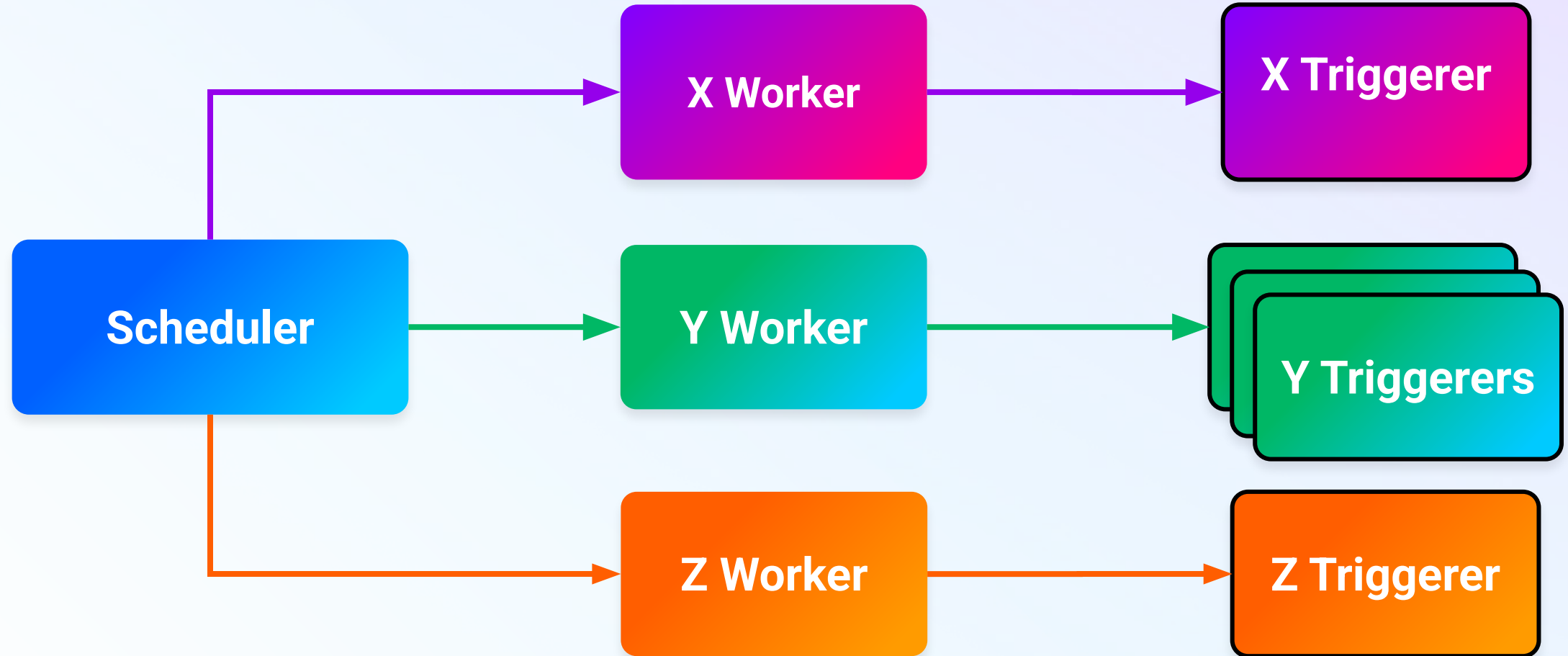


Triggerers “subscribe” to one or more queues, similar to workers

Trigger Queues Visualized



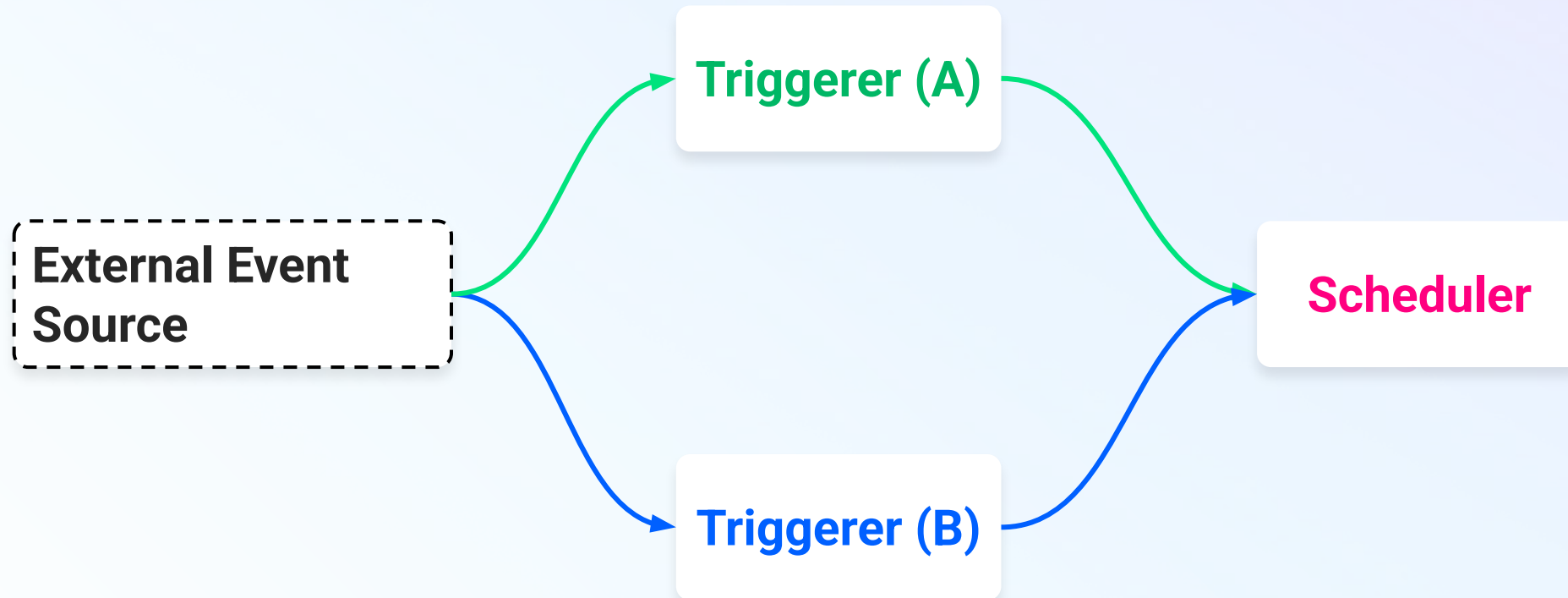
How Datadog Uses Trigger Queues



Support for Event-Based Triggers

Coming Soon!

(Anticipated release: 3.4.0)



*See github.com/apache/airflow/pull/71346

How You Can Use Trigger Queues

For fun and profit!

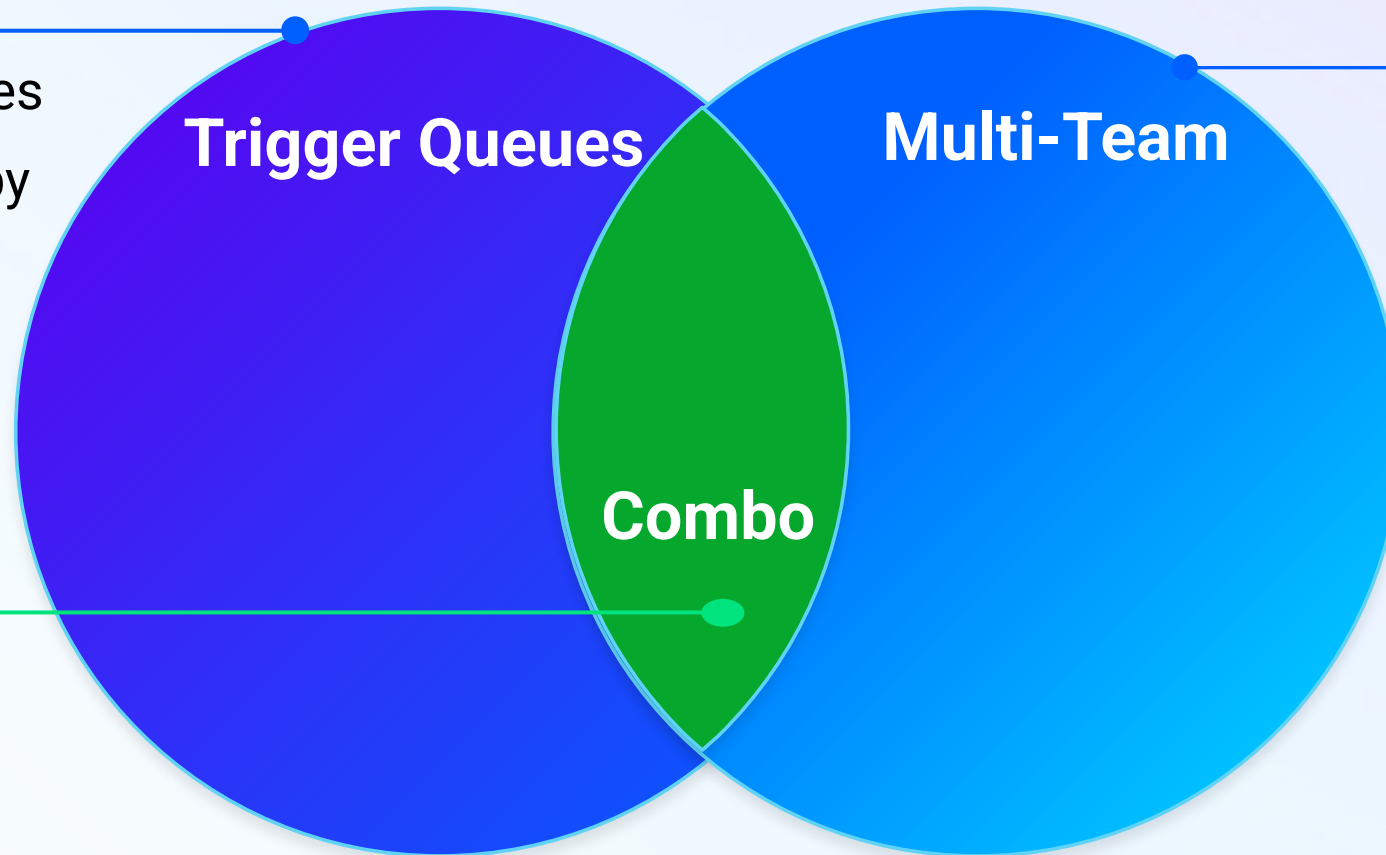
Multi-Team vs. Trigger Queues?

Trigger Queues

- Queue-scoped nodes
- Node assignment by task queue value

Combined

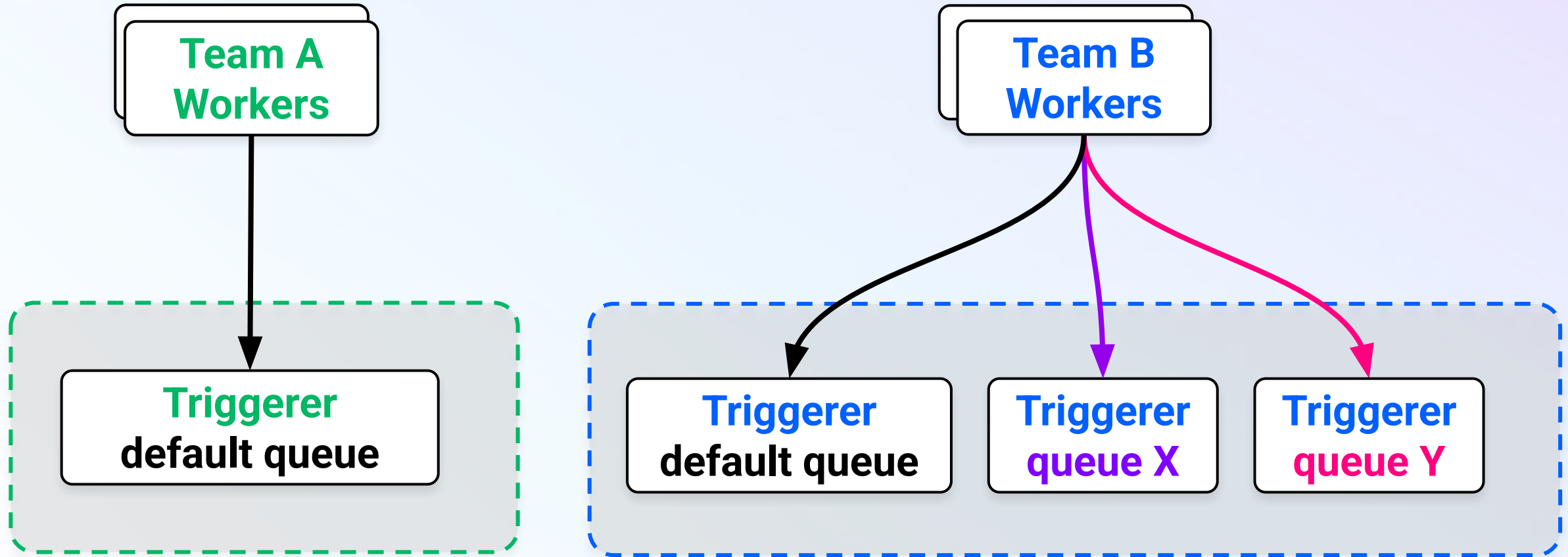
- Distinct, orthogonal trigger assignment controls
- When used together, act as logical **“AND”**



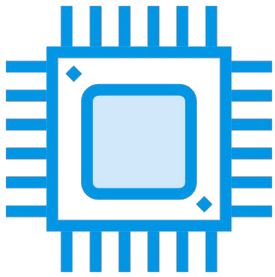
Multi-Team

- Team-scoped node(s)
- Node assignment by DAG origin

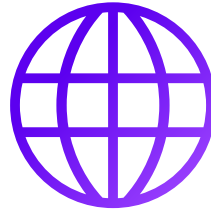
Multi-Team *and* Trigger Queues



Segment your triggers by ...



**Hardware /
architecture**



Spatial locality



Compliance

Thank you

questions?



Scan to view the official
trigger queue docs



DATADOG